Video Transcript

Video 10: Deep Dream Exercise (03:52)

So, I thought it'd be a fun exercise to take a look at a deep neural network and understand a little bit about the layers and what those weights in the neural network represent. So, there's a famous implementation called DeepDream. A gentleman called Alexander Mordvintsev developed and it's implemented here in Google Colabs. And basically what we need to do is we need to modify it so that we can take an image such as this where I feed in myself and then we have it operated on by the deep neural network, and this is how it comes out.

So, I want you to load up one of your images. So, follow the instructions because we need to modify the code, but it should give you an interesting view of yourself as a deep neural net sees you. So, the first thing we need to do is to upload an image of ourselves. So to do that, go here and click on this. And now, we can upload an image here just by drag and drop. So, I'm going to take my image and just drag it onto here. So, once we've loaded the image, we need to copy its path, then scroll down till you find 'URL =' on this line here.

And, we're going to substitute our path that we've just copied for that URL. So, it'll be /content/whatever your image is. Now down below here, we're going to comment out image path. And instead of the image path here, we're just going to use the URL that we've just copied in. So, it's going to pick up our image. So, now you can execute the code and everything should run. Let's go through and just check, so those are all loaded, load this, okay, and it displays the image. So, this is my uploaded image.

Now we need to go through, and now, it's going to actually operate. Here we go, DeepDream model, now the main loop, and calculate the image. So, this is the one I've got. We can alter the step size here on the number of steps, but basically that's DeepDream and how you get very strange images. So, you can play around with that, but I think it gives you some insight into the way that deep learning is operating. Okay, hope you have fun with this. Don't forget to put this up into your portfolio. I think these images are quite interesting. Okay, bye for now.

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